MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 07** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

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Q1. If $\tan\theta = 1$, then the value of $\dfrac{8\sin\theta + 5\cos\theta}{\sin^3\theta - 2\cos^3\theta + 7\cos\theta}$ is: [1]

(A) 2 (B) $2\frac{1}{2}$ (C) 3 (D) $\frac{4}{5}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 2

Justification: $\tan\theta = 1 \implies \theta = 45°$, so $\sin\theta = \cos\theta = \dfrac{1}{\sqrt{2}}$. Evaluating gives $\dfrac{13/\sqrt{2}}{\frac{1}{2\sqrt{2}} - \frac{2}{2\sqrt{2}} + \frac{7}{\sqrt{2}}} = \dfrac{13/\sqrt{2}}{13/(2\sqrt{2})} = 2.$

Q2. The maximum value of $\dfrac{1}{\csc\theta}$ for $0° < \theta \leq 90°$ is: [1]

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\sqrt{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 1

Justification: Since $\dfrac{1}{\csc\theta} = \sin\theta$, on $(0°, 90°]$ its maximum value occurs at $\theta = 90°$, giving $\sin 90° = 1$.

Q3. The value of $(1+\tan^2\theta)(1-\sin\theta)(1+\sin\theta)$ is: [1]

(A) 0 (B) 1 (C) -1 (D) $\sec^2\theta$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 1
Justification: $(1+\tan^2\theta)(1-\sin^2\theta) = \sec^2\theta \cdot \cos^2\theta = 1$.

Q4. The value of $9\sec^2 A - 9\tan^2 A$ is equal to: [1]

(A) 1 (B) 9 (C) 8 (D) 0

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 9
Justification: $9(\sec^2 A - \tan^2 A) = 9(1) = 9$.

Q5. $(1+\tan\theta+\sec\theta)(1+\cot\theta-\csc\theta)$ is equal to: [1]

(A) 0 (B) 1 (C) 2 (D) -1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C) 2
Justification: Converting to $\sin\theta, \cos\theta$:
$$\left(\frac{\cos\theta+\sin\theta+1}{\cos\theta}\right)\left(\frac{\sin\theta+\cos\theta-1}{\sin\theta}\right) = \frac{(\sin\theta+\cos\theta)^2-1}{\sin\theta\cos\theta} = \frac{2\sin\theta\cos\theta}{\sin\theta\cos\theta} = 2.$$

Q6. **Assertion (A):** The value of $\tan 60^\circ \cot 60^\circ = 1$.
Reason (R): $\tan\theta$ and $\cot\theta$ are reciprocals of each other for any valid acute angle θ. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

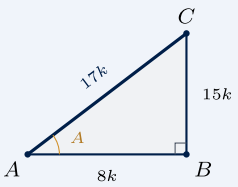
▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A)
Justification: Since $\cot\theta = \dfrac{1}{\tan\theta}$ for $0^\circ < \theta < 90^\circ$, $\tan 60^\circ \cot 60^\circ = 1$. Both are true and (R) explains (A).

Q7. If $15\cot A = 8$, find the values of $\sin A$ and $\sec A$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Find Sides of Right $\triangle ABC$ [1 Mark]

Given $\cot A = \dfrac{AB}{BC} = \dfrac{8}{15}$ ($\angle B = 90°$). Let $AB = 8k$, $BC = 15k$ ($k > 0$).



$$AC = \sqrt{(8k)^2 + (15k)^2} = \sqrt{64k^2 + 225k^2} = \sqrt{289k^2} = 17k$$

Step 2: Evaluate $\sin A$ and $\sec A$ [1 Mark]

$$\sin A = \frac{BC}{AC} = \frac{15}{17}, \qquad \sec A = \frac{AC}{AB} = \frac{17}{8}$$

$$\boxed{\sin A = \mathbf{\frac{15}{17}}, \qquad \sec A = \mathbf{\frac{17}{8}}}$$

Q8. Evaluate: $2\tan^2 45° + \cos^2 30° - \sin^2 60°$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Substitute Standard Values [1 Mark]

Using $\tan 45° = 1$, $\cos 30° = \dfrac{\sqrt{3}}{2}$, and $\sin 60° = \dfrac{\sqrt{3}}{2}$:

$$2(1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2$$

Step 2: Simplify [1 Mark]

$$= 2 + \frac{3}{4} - \frac{3}{4} = \boxed{\mathbf{2}}$$

Q9. If $a\cos\theta - b\sin\theta = c$, prove that $a\sin\theta + b\cos\theta = \pm\sqrt{a^2 + b^2 - c^2}$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Square Given Equation & Desired Expression [1.5 Marks]

Let $x = a\sin\theta + b\cos\theta$. Squaring $a\cos\theta - b\sin\theta = c$ and x and adding:

$$(a\cos\theta - b\sin\theta)^2 + (a\sin\theta + b\cos\theta)^2 = c^2 + x^2$$

Step 2: Expand & Apply $\sin^2\theta + \cos^2\theta = 1$ [1.5 Marks]

$$a^2(\cos^2\theta + \sin^2\theta) + b^2(\sin^2\theta + \cos^2\theta) - 2ab\sin\theta\cos\theta + 2ab\sin\theta\cos\theta = c^2 + x^2$$

$$a^2 + b^2 = c^2 + x^2 \implies x^2 = a^2 + b^2 - c^2 \implies \boxed{\mathbf{a}\sin\theta + \mathbf{b}\cos\theta = \pm\sqrt{\mathbf{a^2 + b^2 - c^2}}} \quad \text{[Hence Proved]}$$

Q10. If $\csc\theta + \cot\theta = q$, prove that $\cos\theta = \dfrac{q^2-1}{q^2+1}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Find $\csc\theta - \cot\theta$ [1 Mark]

$$\csc^2\theta - \cot^2\theta = 1 \implies (\csc\theta+\cot\theta)(\csc\theta-\cot\theta) = 1 \implies \csc\theta - \cot\theta = \frac{1}{q}$$

Step 2: Solve for $\csc\theta$ and $\cot\theta$ [1.5 Marks]

Adding and subtracting the two equations yield $2\csc\theta = q + \frac{1}{q} = \frac{q^2+1}{q}$ and $2\cot\theta = q - \frac{1}{q} = \frac{q^2-1}{q}$.

Step 3: Ratio for $\cos\theta$ [0.5 Mark]

$$\cos\theta = \frac{\cot\theta}{\csc\theta} = \frac{2\cot\theta}{2\csc\theta} = \frac{(q^2-1)/q}{(q^2+1)/q} = \boxed{\mathbf{\frac{q^2-1}{q^2+1}} \quad \text{[Hence Proved]}}$$

SECTION D: Long Answer / HOTS

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. Prove that: $\dfrac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} = \dfrac{1}{\sec\theta - \tan\theta}$, using the identity $\sec^2\theta = 1 + \tan^2\theta$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Divide Numerator & Denominator by $\cos\theta$ [1.5 Marks]

$$\text{LHS} = \frac{\frac{\sin\theta}{\cos\theta} - \frac{\cos\theta}{\cos\theta} + \frac{1}{\cos\theta}}{\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\cos\theta} - \frac{1}{\cos\theta}} = \frac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1}$$

Step 2: Substitute $1 = \sec^2\theta - \tan^2\theta$ in Denominator [2 Marks]

$$\text{Den} = (\tan\theta - \sec\theta) + (\sec^2\theta - \tan^2\theta) = -(\sec\theta - \tan\theta) + (\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$$

$$\text{Den} = (\sec\theta - \tan\theta)(\sec\theta + \tan\theta - 1)$$

Step 3: Cancel Common Factor $(\tan\theta + \sec\theta - 1)$ [1.5 Marks]

$$\text{LHS} = \frac{\tan\theta + \sec\theta - 1}{(\sec\theta - \tan\theta)(\tan\theta + \sec\theta - 1)} = \boxed{\frac{\mathbf{1}}{\sec\theta - \tan\theta} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q12. If $a\cos\theta + b\sin\theta = m$ and $a\sin\theta - b\cos\theta = n$, prove that $a^2 + b^2 = m^2 + n^2$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Square Both Equations [2 Marks]

$$m^2 = (a\cos\theta + b\sin\theta)^2 = a^2\cos^2\theta + b^2\sin^2\theta + 2ab\sin\theta\cos\theta$$

$$n^2 = (a\sin\theta - b\cos\theta)^2 = a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta$$

Step 2: Add m^2 and n^2 [2 Marks]

$$m^2 + n^2 = a^2(\cos^2\theta + \sin^2\theta) + b^2(\sin^2\theta + \cos^2\theta) + 2ab\sin\theta\cos\theta - 2ab\sin\theta\cos\theta$$

Step 3: Apply $\sin^2\theta + \cos^2\theta = 1$ [1 Mark]

$$m^2 + n^2 = a^2(1) + b^2(1) \implies \boxed{\mathbf{a^2 + b^2 = m^2 + n^2} \quad \text{[Hence Proved]}}$$

Q13. Fire Brigade Aerial Turntable Ladder Rescue Operation

A municipal fire rescue truck carries an aerial turntable ladder. During a rescue drill, the ladder base is mounted on the truck platform at point P. The ladder PR of maximum extension length 20 m is inclined at an angle of $\theta = 60^\circ$ to the horizontal to reach a window ledge R of a burning building. **Based on the above geometric model, answer the following questions:**

(i) Calculate the vertical reach height RQ of the ladder above the truck platform level. [1]

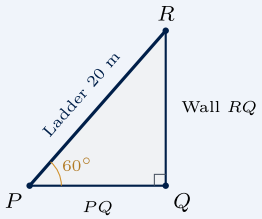
(ii) Find the horizontal distance PQ from the ladder turntable to the building wall. [1]

(iii) If safety regulations permit a maximum inclination angle of 45° in high winds, what would be the vertical reach height of the ladder at 45° extension of 20 m? [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Vertical Reach RQ [1 Mark]

$$RQ = PR\sin 60^\circ = 20 \times \frac{\sqrt{3}}{2} = \boxed{\mathbf{10\sqrt{3}}\text{ m} \approx 17.32\text{ m}}$$



Sub-part (ii): Horizontal Distance PQ [1 Mark]

$$PQ = PR\cos 60^\circ = 20 \times \frac{1}{2} \implies \boxed{PQ = \mathbf{10}\text{ m}}$$

Sub-part (iii): Vertical Reach at 45° Inclination [2 Marks]

$$RQ' = 20\sin 45^\circ = 20 \times \frac{1}{\sqrt{2}} = \frac{20\sqrt{2}}{2} = \boxed{\mathbf{10\sqrt{2}}\text{ m} \approx 14.14\text{ m}}$$

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